

Status	Finished
Started	Wednesday, 11 March 2026, 2:17 PM
Completed	Wednesday, 11 March 2026, 3:35 PM
Duration	1 hour 17 mins
Grade	40.00 out of 40.00 (100%)

Question 1

Correct

Mark 15.00 out of 15.00

Coordinate frame 1 is rotated by 90° **anticlockwise** around the **y**-axis and displaced by $(-2, -1, 0)$ with respect to coordinate frame 0.

Write the homogeneous transformation matrix T_1^0 . Give numerical answers correct to 3 decimal places.

 $T_1^0 =$

0	0	1	-2
0	1	0	-1
-1	0	0	0
0	0	0	1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 0 & 0 & 1 & -2 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

✓ Correct answer, well done.

Part B: Point $P_1 = [-1, -1, 0]^T$. What is P_0 ? Enter as $[x, y, z]$

 $P_0 = [-2, -2, 1]$

Your last answer was interpreted as follows:

$$[-2, -2, 1]$$

✓ Correct answer, well done.

Coordinate frame 2 is rotated by 45° **clockwise** around the **z**-axis and displaced by $(0, -1, 1)$ with respect to frame 1.

Part C: Write the homogeneous transformation matrix T_2^1 .

 $T_2^1 =$

0.707	0.707	0	0
-0.707	0.707	0	-1
0	0	1	1
0	0	0	1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 0.707 & 0.707 & 0 & 0 \\ -0.707 & 0.707 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

✓ Correct answer, well done.

Part D: Calculate T_2^0 .

 $T_2^0 =$

0	0	1	-1
-0.707	0.707	0	-2
-0.707	-0.707	0	0
0	0	0	1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 0 & 0 & 1 & -1 \\ -0.707 & 0.707 & 0 & -2 \\ -0.707 & -0.707 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

✓ Correct answer, well done.

<p>Part A: \[T^0_1 = \left(\begin{array}{cccc} 0 & 0 & 1 & -2 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)\]</p>

<p>Part B: \[P_0 = \left(\begin{array}{c} -2 \\ -2 \\ 1 \end{array}\right)\]</p>

<p>Part C: \[T^1_2 = \left(\begin{array}{cccc} 0.707106781187 & 0.707106781187 & 0 & 0 \\ -0.707106781187 & 0.707106781187 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right)\]</p>

<p>Part D: $T^{0_2} = T^{0_1} \cdot T^{1_2} = \begin{pmatrix} 0 & 0 & 1 & -2 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ </p>

The answer $\begin{bmatrix} 0 & 0 & 1 & -2 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ would be correct.

The answer $[-2, -2, 1]$, which can be typed as **$[-2, -2, 1]$** , would be correct.

The answer $\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ would be correct.

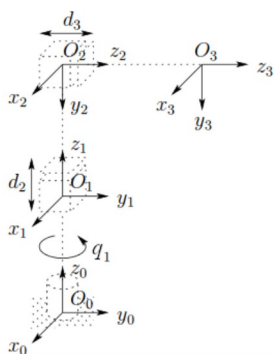
The answer $\begin{bmatrix} 0 & 0 & 1 & -1 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & -2 \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ would be correct.

Question 2

Correct

Mark 5.00 out of 5.00

The RPP arm used for this question is shown below. Note that the first 2 joints are in the same place (The distance between them is 0)



Fill in the corresponding table of DH parameters (using degrees, not radians)

i	a_i	α_i	d_i	θ_i
1	0 ✓	0 ✓	0 ✓	q1 ✓
2	0 ✓	-90 ✓	d2 ✓	0 ✓
3	0 ✓	0 ✓	d3 ✓	0 ✓

Question 3

Correct

Mark 10.00 out of 10.00

The Denavit-Hartenberg parameters for a two-link robot are:

i	a_i	α_i	d_i	θ_i
1	1	0	1	θ_1
2	1	0	1	θ_2

Using the **standard Denavit-Hartenberg convention**, compute the homogeneous transformation matrices

Enter the matrix elements below.

Use **theta1** and **theta2** for the joint variables. Note that you can write expressions like $\cos(-\pi/2)\sin(\theta_1)$ which will be rendered as $\cos(-\pi/2)\sin(\theta_1)$

$T_1^2 =$

cos(theta2)	-sin(theta2)	0	cos(theta2)
sin(theta2)	cos(theta2)	0	sin(theta2)
0	0	1	1
0	0	0	1

Your last answer was interpreted as follows:

$$\begin{bmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 & \cos(\theta_2) \\ \sin(\theta_2) & \cos(\theta_2) & 0 & \sin(\theta_2) \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The variables found in your answer were: $[\theta_2]$

$T_0^1 =$

cos(theta1)	-sin(theta1)	0	cos(theta1)
sin(theta1)	cos(theta1)	0	sin(theta1)
0	0	1	1
0	0	0	1

Your last answer was interpreted as follows:

$$\begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & \cos(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) & 0 & \sin(\theta_1) \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The variables found in your answer were: $[\theta_1]$

$T_0^2 =$

cos(theta1+theta2)	-sin(theta1+theta2)	0	cos(theta1)+cos(theta2)
sin(theta1+theta2)	cos(theta1+theta2)	0	sin(theta1)+sin(theta2)
0	0	1	2
0	0	0	1

Your last answer was interpreted as follows:

$$\begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 & \cos(\theta_1) + \cos(\theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 & \sin(\theta_1) + \sin(\theta_2) \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The variables found in your answer were: $[\theta_1, \theta_2]$

✓ Correct answer, well done.

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The answer $\begin{bmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 & \cos(\theta_2) \\ \sin(\theta_2) & \cos(\theta_2) & 0 & \sin(\theta_2) \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ would be correct.

The answer $\begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & \cos(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) & 0 & \sin(\theta_1) \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ would be correct.

The answer $\begin{bmatrix} \cos(\theta_1) \cdot \cos(\theta_2) - \sin(\theta_1) \cdot \sin(\theta_2) & -\cos(\theta_1) \cdot \sin(\theta_2) - \sin(\theta_1) \cdot \cos(\theta_2) & 0 & -\sin(\theta_1) \cdot \sin(\theta_2) + \cos(\theta_1) \cdot \cos(\theta_2) + \cos(\theta_1) \\ \cos(\theta_1) \cdot \sin(\theta_2) + \sin(\theta_1) \cdot \cos(\theta_2) & \cos(\theta_1) \cdot \cos(\theta_2) - \sin(\theta_1) \cdot \sin(\theta_2) & 0 & \cos(\theta_1) \cdot \sin(\theta_2) + \sin(\theta_1) \cdot \cos(\theta_2) + \sin(\theta_1) \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ would be correct.

Question 4

Correct

Mark 10.00 out of 10.00

You have an RPP arm with the following transformation matrices.

Work out J_ω and J_v .

$$T_1^0 = \begin{pmatrix} \cos(q_1) & -\sin(q_1) & 0 & 0 \\ \sin(q_1) & \cos(q_1) & 0 & 0 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$T_2^0 = \begin{pmatrix} \cos(q_1) & 0 & -\sin(q_1) & 0 \\ \sin(q_1) & 0 & \cos(q_1) & 0 \\ 0 & -1 & 0 & 5 + q_2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$T_3^0 = \begin{pmatrix} \cos(q_1) & 0 & -\sin(q_1) & -\sin(q_1)q_3 \\ \sin(q_1) & 0 & \cos(q_1) & \cos(q_1)q_3 \\ 0 & -1 & 0 & 5 + q_2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Note: $d_1 = 5$ m is a fixed structural parameter (not a joint variable). Also note that you can use expressions like $\sin(q_1)\cos(q_3)$ which will be rendered as $\sin(q_1)\cos(q_3)$

$J_\omega =$

0	0	0
0	0	0
1	0	0

Your last answer was interpreted as follows:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

✓ Correct answer, well done.

Use exact symbolic expressions in terms of q_1, q_2, q_3 .

$J_v =$

-q3cos(q1)	0	-sin(q1)
-q3sin(q1)	0	cos(q1)
0	1	0

Your last answer was interpreted as follows:

$$\begin{bmatrix} (-q_3) \cdot \cos(q_1) & 0 & -\sin(q_1) \\ (-q_3) \cdot \sin(q_1) & 0 & \cos(q_1) \\ 0 & 1 & 0 \end{bmatrix}$$

The variables found in your answer were: $[q_1, q_3]$

✓ Correct answer, well done.

Linear velocity of the endpoint

Given:

Joint 1: $q_1 = 2\pi/5$ rad, $\dot{q}_1 = 1$ rad/s

Joint 2: $q_2 = 1$ m, $\dot{q}_2 = 2$ m/s

Joint 3: $q_3 = 3$ m, $\dot{q}_3 = 3$ m/s

Compute the linear velocity vector

Enter the 3×1 vector. Note that you must not use symbols in this vector, and your answer should be correct to 3 decimal places.

$\mathbf{v} =$

-3.78
-1.926
2

Your last answer was interpreted as follows:

$$\begin{bmatrix} -3.78 \\ -1.926 \\ 2 \end{bmatrix}$$

✓ Correct answer, well done.

The answer $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ would be correct.

The answer $\begin{bmatrix} -\cos(q_1) \cdot q_3 & 0 & -\sin(q_1) \\ -\sin(q_1) \cdot q_3 & 0 & \cos(q_1) \\ 0 & 1 & 0 \end{bmatrix}$ would be correct.

The answer $\begin{bmatrix} -3 \cdot \sin\left(\frac{2\pi}{5}\right) - 3 \cdot \cos\left(\frac{2\pi}{5}\right) \\ 3 \cdot \cos\left(\frac{2\pi}{5}\right) - 3 \cdot \sin\left(\frac{2\pi}{5}\right) \\ 2 \end{bmatrix}$ would be correct.